Chapter 2
Networking

Introduction To Networking

With the advanced networking facilities of the IBM AS/400 you can use the IBM System Network Architecture Distribution Services (SNADS) to distribute objects (folders, libraries, and so on) and messages to other systems linked in an Advanced Peer-To-Peer Communications (APPC) or Advanced Peer-To-Peer Networking (APPN) network.

Network Manager offers a user-friendly method of configuring and maintaining your network.

System Network Architecture Distribution Services (SNADS)

SNADS is the IBM SNA distribution service used by the networking utilities to distribute objects and messages to other systems linked directly or indirectly to your system.

To enable SNADS distribution throughout your network you must configure the two main elements of SNADS on every system within the network. These are:

- Distribution queues
- Routing table.

Distribution Queues

Distribution queues are used to send distributions to other systems in your network. The distribution queue contains, for each configured queue, the name of the remote location to which the queue should be sent and the queue type i.e. SNADS.

In an APPC network you must configure distribution queue entries for each system linked directly to the local system. In an APPN network you should configure distribution queue entries for all the remote systems in your system regardless of whether they are adjacent systems or not.

Routing Table

The routing table contains entries for each remote system in your network. Each routing table entry has the name of the distribution queue from which distributions to the remote system can be sent. This information determines which route is used to send a distribution from one system to another.

In an APPN network you generally configure both distribution queues and routing entries for every system in your network. Each routing entry specifies the unique distribution queue for each remote system.

APPC networks support generic routing whereby you can specify *ANY as the target system provided that all systems linked directly to your system contain a full routing table of all possible distribution systems, or a *ANY entry to route to another system with a complete routing table of all possible distribution systems.

Secondary System Name

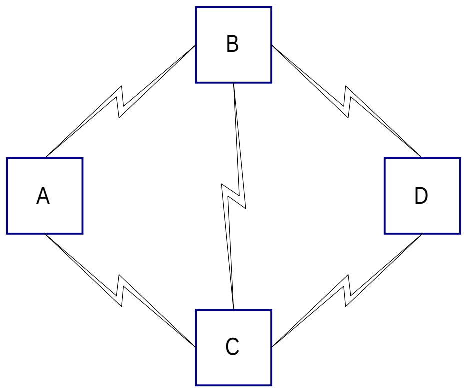
Use the third option in distribution services to define an alternative name for your local system. Distributions can then be routed to your local system using this alternative name.

In Network Manager this facility can be used to control multiple spool file distributions to your system, permitting output from a remote system to be directed to output queues and printers locally i.e. for each department. This would enable you to separate sensitive documents from your normal network spool files.

APPC And APPN

You can use Network Manager to configure your network as either an Advanced Peer-to-Peer Communications (APPC) or an Advanced Peer-To-Peer Networking (APPN) network. The method of distributing objects between systems and the configuration of those systems differ depending on whether you configure your network as APPC or APPN.

For example, in the following simple network you may wish to send a distribution from system A to system D.



In an APPC network the distribution is sent point-to-point:

From a SNADS job on system A to a SNADS job on system B then from a SNADS job on system B to a SNADS job on system D.

This method of intermediate routing involves a SNADS session being started on each intermediate system.

In an APPN network APPN sessions can be started directly between all systems in the network. SNADS does no intermediate routing and system A can be viewed as being linked to every other SNADS system in the network. So to transmit a distribution from system A to system D a SNADS job on system A starts an APPN session between system A and system D through system B.

An APPN network has the advantage that no SNADS resources are used on the intermediate systems.

APPN networks also support alternative routing. For example, if the communications line between systems B and D becomes inoperable, SNADS attempts to start a new APPN session via another route (through system C). When the user requests a new SNADS session to re-send the distribution SNADS first attempts to use the original route, and then switch to the alternative route if this is still inoperable. SNADS continues to monitor the communications lines and return to using the original route when the communications line between systems B and D becomes operational.

SNA Distribution Services Configuration

So you can distribute objects throughout your network you must configure IBM's SNADS system so that it can identify the other systems in your network.

It is assumed that suitable lines, controllers, and devices have already been configured and are working. It is assumed that pass-through is also working.

Note: It is necessary to be signed on as QSECOFR for all the following operations.

QSNADS Subsystem

Before configuring SNADS make sure that the subsystem QSNADS is not running. To check whether QSNADS is active enter the command:

WRKACTJOBSBS(QSNADS)

If any active jobs are displayed enter the command:

ENDSBSSBS(QSNADS)OPTION(*IMMED)

Select **F5** to refresh the screen until all active jobs are ended and the message No active jobs to display is displayed.

Note: Terminating subsystem QSNADS delays those jobs currently transmitting. SNADS automatically re-transmits these jobs when the subsystem QSNADS is re-activated. However should any lengthy jobs be currently active, you should wait until this job is finished before ending the subsystem. Use the WRKDSTQ command if necessary to check the status of current send requests.

Configure Distribution Queues Screen

After ending the QSNADS subsystem, enter the command CFGDSTSRV. Select option 1 and press **Enter**.

This displays a list of existing distribution queues.

```
                          Configure Distribution Queues

 Type options, press Enter.
   2=Change   4=Delete   5=Display details

                                       Remote                        Remote
 Opt  Queue Name          Queue Type    Location Name  Mode Name     Net ID
      ALLIANCE             *SNADS        S44A1957       AGENT        *LOC
  _   AMTER1               *SNADS        AMTER001       AGENT        *LOC
  _   AMTER2               *SNADS        AMTER002       AGENT        *LOC
  _   BLOHORN              *SNADS        BLOHORN        AGENT        *LOC
  _   BRISB30A             *SNADS        BRISB30A       PRINT        *LOC
  _   BRISTOOL             *SNADS        BRISTOOL       PRINT        *LOC
  _   CALB30A              *SNADS        CALB30A        PRINT        *LOC
  _   CAMB400A             *SNADS        CAMB400A       PRINT        *LOC
  _   CEPB30               *SNADS        CEPB30         AGENT        *LOC
  _   CHER400A             *SNADS        CHER400A       TRANSFER     *LOC
  _   CHIC400              *SNADS        CHIC400        TRANSFER     *LOC
  _   CHWDB10A             *SNADS        CHWDB10A       *NETATR      *LOC
  _   CHWDB40A             *SNADS        CHWDB40A       PRINT        *LOC
  _   CHWD38               *SNADS        CHWD38         PRINT        *LOC           +
  _

 F3=Exit       F5=Refresh          F6=Add distribution queue
 F10=Work with distribution queues                         F12=Cancel
```

Fields:

Opt

Selection field. Enter one of the following:

2 - Change. To amend the selected distribution queue.

4 - Delete. To delete the selected distribution queue.

5- Display details. To display the details for the selected distribution queue.

To add a new distribution queue select **F6=Add Distribution Queue**.

Add Distribution Queue Screen (F6)

To add a new distribution queue select **F6=Add Distribution Queue** on the Configure Distribution Queue Screen.

Use this screen to enter the details for the distribution queue.

```
                       Add Distribution Queue                 Page 1 of 2

 Type choices, press Enter.

   Queue . . . . . . . . . . .    NEWSYS           Name
   Queue type . . . . . . . . .   *SNADS           *SNADS, *RPDS, *DLS
   Remote location name . . . .   NEWSYS           Name
   Mode . . . . . . . . . . . .   *NETATR          Name, *NETATR
   Remote net ID . . . . . . .    *LOC             Name, *LOC, *NONE
   Local location name . . . .    *LOC             Name, *LOC
   Normal priority:
     Send time:
       From/To . . . . . . . .    __ : __  __ : __  00:00-23:59
       Force . . . . . . . . .    __ : __           00:00-23:59
     Send depth . . . . . . . .   __1               1-999, blank
   High priority:
     Send time:
       From/To . . . . . . . .    __ : __  __ : __  00:00-23:59
       Force . . . . . . . . .    __ : __           00:00-23:59
     Send depth . . . . . . . .   __1               1-999, blank
                                                                    More...
 F3=Exit     F12=Cancel
```

Fields:

Queue

Enter the name of the queue in which distributions are stored before being sent. It is recommended that the name of the queue is the same as the name of the target system.

Queue Type

Enter the type of distribution queue. For a SNADS network this must be *SNADS.

Remote Location Name

Enter the name of the system to which distributions are sent to users not on your system. This should be the name of the default local location name on the remote system (use the command DSPNETA on the remote system), or the name of the remote location name of the communications device for the target system.

Mode

If you have defined modes to optimise lines, enter one here. You can use the default *NETATR (the mode in network attributes), but this can significantly affect pass-through performance.

Note: You can define modes for both batch and interactive communications. Set up batch modes with lower values for pacing and request unit *size. If the same mode is used for both batch and interactive communications then batch transmissions will significantly impact interactive pass-through.*

Remote Net ID

Enter the remote network ID to which your distributions are sent. Enter *LOC so the system can determine which value to use.

Local Location Name

Enter the name that identifies your system to remote systems in the network. This name must match the remote location name specified in the distribution queue of remote systems. Enter *LOC so the system can determine which value to use.

<u>Normal Priority:</u>

The normal priority portion of the queue is those distributions having a service level of data low.

Note: For a distribution having a service level of data low the object size of the distribution is limited to 2 gigabytes.

<u>High Priority:</u>

The high priority portion of the queue is for those distributions having a service level of fast, status, or data high. If you use force time it should be earlier than the force time on the normal priority queue.

Note: For distributions with a service level of fast or status the object size of the distribution is limited to 4k. For distributions having a service level of data high the object size of the distribution is limited to 16 megabytes.

Note: Distributions are normally scheduled to the normal priority portion of the queue but you can move them between the normal and high priority portions using the command WRKDSTQ. When entries exist on both queues for the same target system, the high priority entries are sent first.

Send Time: From/To

Enter the start and finish time for transmissions of this particular queue. If you do not enter a time the transmissions are controlled by send depth.

Send Time: Force

Enter a specific time of day to force the transmission of all the distributions in the queue regardless of the send depth. If there are from or to times entered, the force time must be set for during this period.

Send Depth

Enter a value from 1 to 999 to specify the number of distributions required on the queue before transmission can commence. If a time period has not been entered then this value controls transmission. If both a force time and send depth are entered the force time dictates the transmission time.

Note: If you do not enter any send or force times and leave the send depth field blank, all transmissions must be started by the System Operator using one of the following:

Option 2 (send queue) on the Work with Distribution Queues display.

(Work with Distribution Queue command (WRKDSTQ).

Send Distribution Queue command (SNDDSTQ).

Number Of Retries

Enter the number of times a SNADS sender attempts to send distributions from a SNADS distribution queue after failure occurs. Enter 0 to attempt no retries.

Number Of Minutes Between Retries

Enter the time in minutes between retry attempts. Enter 0 to specify the SNADS sender does not wait before attempting to re-send the distributions.

Ignore Time/Depth Values While Receiving Send Queue

Enter whether, when a SNADS receiver becomes active, a SNADS sender is started on the same connection. If Send Queue is set to Yes, and if a SNADS receiver becomes active using the same configured remote location name, mode name, local location name, and remote network ID as the distribution queue, the time and depth limitations for a queue are ignored and all the queued distributions are sent.

Tip: Leave this field set to the default No if you do not want the SNADS receiver to affect the sending of any distributions from this queue.

Press **Enter** to configure the distribution queue.

Configure Routing Table Screen

After ending the QSNADS subsystem, enter the command CFGDSTSRV. Select option 2 and press **Enter**.

The routing table specifies the distribution queue used to send a distribution to a specific target system.

A distribution is routed by IBM AS/400 SNADS based on its service level (fast, status, data high/low) and its destination. You must configure a routing table entry for all system names to which you want to send distributions, you must also have an entry for each service level used.

Note: You must configure the status service level since this is used by SNADS to report any error feedback.

```
                              Configure Routing Table

 Type options, press Enter.
   2=Change   4=Delete   5=Display details

              -------System--------
 Opt    Name        Group           Description
  _     AMTER1                      Amter Ltd  AS/400 B60.
  _     AMTER2                      Amter Ltd  AS/400 B45.
  _     AS400A                      The Computing Group AS/400
  _     BLOHORN                     Unilever Blowhorn AS/400 Ivory Coast
  _     BRISB30A                    Bristol AS/400 Model B30
  _     BRISTOOL                    JBA Bristol Development C04.
  _     CALB30A                     Calthorpe Rd AS/400 Model B30
  _     CAMB400A                    JBA Cambridge AS/400.
  _     CAP400                      AS/400 Eniac Capelle
  _     CEPB30                      C.E.P Limited AS/400
  _     CHER400A                    JBA Chertsey AS/400 B40.
  _     CHIC400                     JBA Incorporated Chicago AS/400
  _     CHWDB10A                    JBA Chorleywood Demonstration AS/400 B10.
  _     CHWDB40A                    Chorley Wood AS/400 Model B40              +

 F3=Exit      F5=Refresh         F6=Add routing table entry
 F12=Cancel
```

Fields:

Opt

Selection field. Enter one of the following:

2 - Change. To amend the selected routing table.

4 - Delete. To delete the selected routing table.

5- Display details. To display the details for the selected routing table.

To add a routing table entry select **F6=Add Routing Table Entry**.

Add Routing Table Entry (F6)

To add a routing table entry select **F6=Add Routing Table Entry** on the Configure Routing Table Screen.

Use this screen to enter the routing Table details.

```
                         Add Routing Table Entry

 Type choices, press Enter. (At least one queue name is required.)

   System name/Group  . .   NEWSYS
   Description  . . . . .   New System
   Service level:
     Fast:
       Queue name . . . .   NEWSYS          Distribution queue name
       Maximum hops . . .   *DFT            Number of hops' *DFT
     Status:
       Queue name . . . .   NEWSYS
       Maximum hops . . .   *DFT
     Data high:
       Queue name . . . .   NEWSYS
       Maximum hops . . .   *DFT
     Data low:
       Queue name . . . .   NEWSYS
       Maximum hops . . .   *DFT


 F3=Exit      F12=Cancel
```

Fields:

System Name/Group

Enter the remote location name as the system name. System group is not used by the IBM AS/400.

Description

Enter a description of the target system.

Tip: In a simple configuration use the same route for all service levels. For each service level configured you must specify the following:

Queue Name

If the target system is adjacent or you are using APPN, enter the queue previously configured for the target system. If you are using APPC, enter the queue name of an adjacent system which will forward the distributions to the target machine.

Maximum Hops

Enter the maximum number of times you can receive a distribution forward it on the way to its final destination.

To calculate this value, determine the number of intermediate nodes between the source and target systems and add at least one. If the maximum hop count is exceeded before the distribution reaches its destination the transmission is ended. SNADS will notify the sender that an error occurred in the distribution.

If you do not enter a maximum hops value the *DFT value is assigned. This value uses the default maximum hop count value in the network attribute at the time the distribution is being routed. This value is set at 16 when the system is shipped.

Press **Enter** to add the entry to the routing table.

Manage Network

When you have configured the SNADS queues and routing you must then configure the local system and all the remote systems in the network.

You must first start the subsystem QSNADS using the command:

STRSBSSBSD(QSNADS)

You can define each remote system in a similar way to the local system. Each remote system defined on your system requires the creation of a User Profile with the same name as the target system. You are prompted to create this User Profile, along with the relevant queues, and authority granted to the objects to be used. You are also prompted to create the SNADS directory entries.

Note: To configure the required objects for your network you must be signed on as QSECOFR.

*Note: Throughout this procedure, description screens are displayed to tell you what to do next. On these screens, press **Enter** to continue to the maintenance screen.*

Manage Network Configuration Screen 1

Select Manage Network and press **Enter**.

The Manage Network Configuration window displays a window list of available systems.

```
NW301                                                                System: STUDB30A
 Manage Network Configuration
 1=Select, 2=Authorise users    New or existing System Name . . .  STUDGERM
     System   Type          Description
 _   ALF      AS/400 Remote T & G Barcelona Creative Group
 _   ALL      *DSTL  Remote All AS/400s and S/38s
 _   ALL38    *DSTL  Remote All System/38's
 _   ALL400   *DSTL  Remote All AS/400's
 _   BRISB30A AS/400 Remote Bristol AS/400 Model B30
 _   CALB30A  AS/400 Remote Calthorpe Rd AS/400 Model B30
 _   CAMB400A AS/400 Remote JBA Cambridge AS/400
 _   CAMLEE   AS/400 Remote Campbell Lee - Falkirk
 _   CEPB30   AS/400 Remote Remote CEP AS/400
 _   CHER36   S/36   Remote Chertsey System/36
 _   CHER400A AS/400 Remote Chertsey AS/400 A machine
 _   CHER400B AS/400 Remote Chertsey AS/400 B machine
 _   CHICE02  AS/400 Remote Chicago Sales AS/400 JBA International plc
 _   CHIC400  AS/400 Remote Chicago AS/400 JBA International Inc
 _   CHWDB10A AS/400 Remote Chorleywood Demonstration B10 AS/400
 _   CHWDB40A AS/400 Remote Chorleywood AS400 Model B40
 _   CWMB36   S/36   Remote Cwmbran System/36
                                                                        More...
 F3=Exit    F5=Refresh
```

Fields:

New Or Existing System Name

Enter the name of the local system you want to configure.

Untitled

Selection field. Enter one of the following:

1 - Select. To select the system to configure.

2 - Authorise Users. To authorise users to a system.

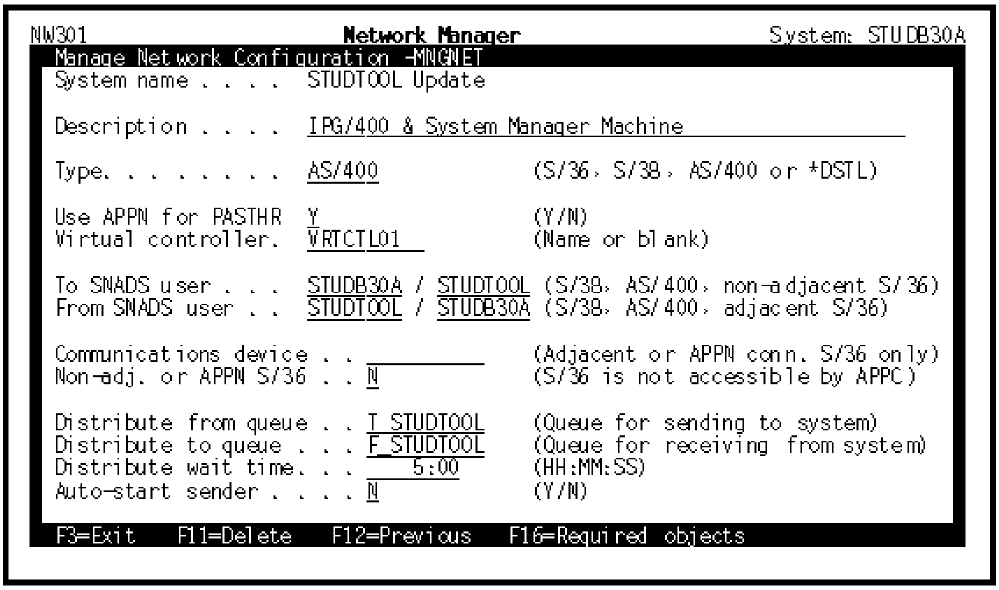
Enter 1 against one of the remote systems in your network and press **Enter** to display the Manage Network Configuration Screen 2.

Manage Network Configuration Screen 2

In the Manage Network Configuration window
select a system with option 1 and press **Enter**.

Use this screen to enter the details for the system.

The local system employs the user profile NETWORK when communicating with remote systems. This user profile must be created, along with relevant queues, and authority granted to the objects to be used.



```
NW301                     Network Manager                  System: STUDB30A
 Manage Network Configuration -MNGNET
 System name . . . .   STUDTOOL Update

 Description . . . .   IPG/400 & System Manager Machine

 Type. . . . . . . .   AS/400             (S/36, S/38, AS/400 or *DSTL)

 Use APPN for PASTHR   Y                  (Y/N)
 Virtual controller.   VRTCTL01           (Name or blank)

 To SNADS user . . .   STUDB30A / STUDTOOL (S/38, AS/400, non-adjacent S/36)
 From SNADS user . .   STUDTOOL / STUDB30A (S/38, AS/400, adjacent S/36)

 Communications device . .                (Adjacent or APPN conn. S/36 only)
 Non-adj. or APPN S/36 . . N              (S/36 is not accessible by APPC)

 Distribute from queue . . T_STUDTOOL     (Queue for sending to system)
 Distribute to queue . . . F_STUDTOOL     (Queue for receiving from system)
 Distribute wait time. . .     5:00       (HH:MM:SS)
 Auto-start sender . . . . N              (Y/N)

 F3=Exit    F11=Delete    F12=Previous    F16=Required objects
```

Fields:

Note: This is a list of all possible fields. The actual fields displayed depend upon the system selected.

Description

Enter a description of the system.

Library For Network System Objects

Enter the library in which you store the configuration objects. This should be a non-IBM library in your library list.

Type

Enter the system type, such as IBM AS/400. For local systems this is automatically completed.

*Note: you can use type *DSTL to configure a list of several previously configured systems to which distributions can be sent simultaneously.*

Tip: Enter the system type and press **Enter** to display the defaults for the other parameters.

Use Appn For Pasthr

Enter Y if APPN is used for this system.

Virtual Controller

Enter the name of a virtual controller if one is known.

Note: The values used in the Use APPN for PASTHR and Virtual controller fields are used by the MNGPASTHR command.

Snads User

It is recommended that you do not change the SNADS user defaults.

Communications Device

If the remote system is a System 36 and is adjacent or uses APPN enter the communications device for the System 36 in this field.

Leave this field blank if the System 36 is non-adjacent and does not use APPN.

Non-Adj. Or Appn S/36

If the System 36 is non-adjacent or uses APPN enter Y in this field.

Note: For all other systems leave the previous two fields as default.

Distribute From Queue

This defaults to T_NEWSYS (where NEWSYS is the remote location name). Enter a new value if you want distributions sent from a different queue.

Distribute To Queue

This will default to F_NEWSYS (where NEWSYS is the remote location name). Enter a new value if you want distributions sent to a different queue.

Distribute Wait Time

Enter the time interval at which the SNADS sender job is activated. For systems which have few spool file distributions give this a higher value. Where many spool file distributions are sent between two systems leave this at the default time of 5 minutes.

Note: For infrequently used systems enter a long wait period when 'auto-start sender' is set to N.

Auto-Start Sender

This defaults to Y and distributes automatic spool file to any remote system configured Y for this parameter. Spool file distribution is activated periodically depending on the wait time specified in the previous field.

Enter N if you do not want automatic spool file distribution. Spool file distribution can be started manually using Spool File Distribution (see the Start section).

Tip: Using manual activation reduces the number of active jobs on your system while automatic activation requires no user intervention or expert knowledge of the system.

Note: At each stage the important values will be pre-filled in order to simplify the task. You need change no values, only confirm that they are correct. See the IBM Control Language manual for a detailed explanation of the commands used.

Note: Configuration objects must be readily available and are therefore placed in the non-IBM system library. The recommended system library OSLSYS is used here.

To create the various objects required for communication with all remote systems select **F16=Required Objects**. If you are updating the record, you see prompts to change these objects.

***Note: F16** is only displayed if you are signed on as QSECOFR.*

Manage Network Configuration Screen 3

To display this screen, selecting **F16=Required Objects** on the Manage Network Configuration Screen 2 window.

This lists the objects you need to create.

```
NW301                    Network Manager                    System: STUDB30A
 Manage Network Configuration


  The following objects are required by the IPG/400 Networking utilities
  for communication with all remote systems.

  User profile NETWORK
  Authority  to CRTSAVF
  Authority  to CHGSYSLIBL

  Output queue  OSLSYS     /NETWORK
  Message queue OSLSYS     /NETWORK


  Press Enter and you will be prompted to create / change these objects.




 F3=Exit    F12=Previous
```

Press **Enter** to display the Manage Network Configuration Screen 4.

Manage Network Configuration Screen 4

To display this screen, press **Enter** to display on the Manage Network Configuration Screen 3.

This screen displays the user profiles you need to create. For local system this is NETWORK, for remote systems, this is the same name as the remote system.

```
NW301                          Network Manager                     System: STUDB30A
 Manage Network Configuration


  The following parameters of the user profile  NETWORK are important:

  Password *NONE since there is no need to sign-on as this user.

  Job description OSLSYS      /NETWORK
  Message queue   OSLSYS      /NETWORK
  Output queue    OSLSYS      /NETWORK

  Group profile   QPGMR       These values will ensure sufficient authority
  Owner           *GRPPRF     to most of the commands needed by the utility
                              programs.  If your security requirements are
                              strict, you may need to review these values.




 F3=Exit    F12=Previous
```

Press **Enter** to create or change a user profile as appropriate. The Change User Profile screen is displayed.

Change User Profile Screen

To display the screen, press **Enter** on the Manage Network Configuration Screen 4.

This screen displays the default details for the user profile which you must create or change.

```
                         Change User Profile (CHGUSRPRF)

Type choices, press Enter.

User profile . . . . . . . . . . USRPRF     > NETWORK
User password  . . . . . . . . . PASSWORD     *SAME
User class . . . . . . . . . . . USRCLS       *USER
Current library  . . . . . . . . CURLIB       *CRTDFT
Initial program to call  . . . . INLPGM       *NONE
  Library  . . . . . . . . . . .
Initial menu . . . . . . . . . . INLMNU       MAIN
  Library  . . . . . . . . . . .                *LIBL
Limit capabilities . . . . . . . LMTCPB       *NO
Text 'description' . . . . . . . TEXT         'Network utility local user.'


                           Additional Parameters

Job description  . . . . . . . . JOBD       > NETWORK
  Library  . . . . . . . . . . .            >   OSLSYS
                                                                   More...
F3=Exit   F4=Prompt   F5=Refresh   F10=Additional parameters   F12=Cancel
F13=How to use this display        F24=More keys
```

to transfer reports from one system to another, you must create two output queues. One queue is the target system queue when transferring spool files from the local system to a remote system. This queue is given the name of the remote system prefixed with T_. The second queue is the receiving queue when a spool file is transferred from a remote system to the local system. This queue is given the name of the remote system prefixed with F_.

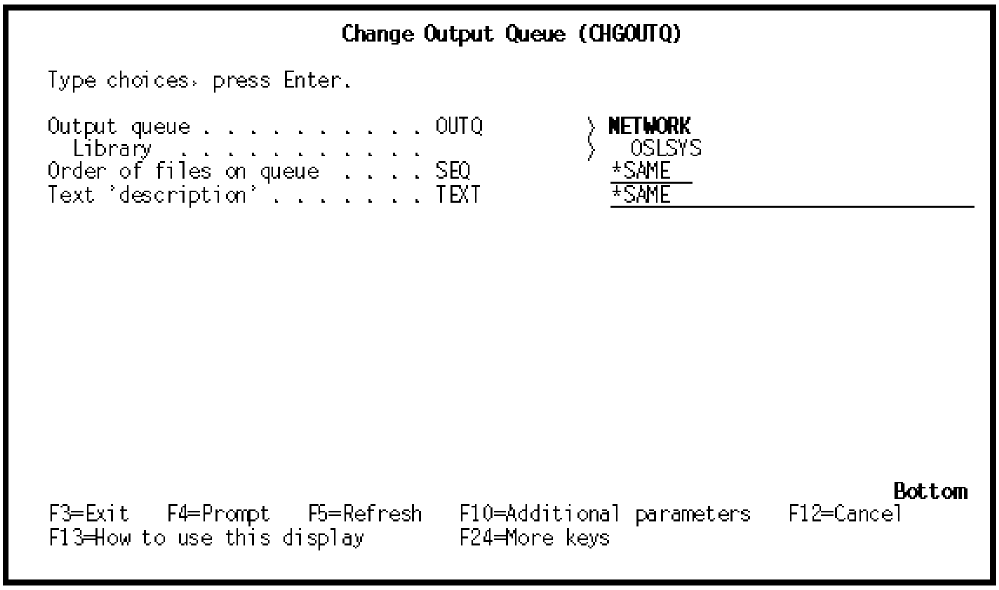
The target output queue for transferring prints to the remote system must be created using the Manage Network Configuration screen.

Press **Enter** twice to display the Change Output Queue Screen.

Change Output Queue Screen

To display this screen, press **Enter** to display the description screen, then press **Enter** again.

Use this screen to create or change the output queue.



Press **Enter** to display the description screen, then press **Enter** again to display the Change job Configuration Screen.

Change Job Configuration

To display this screen, press **Enter** to display the description screen, then press **Enter** again.

Use this screen to create or change the job description.

Batch jobs require a job description. For remote systems job description have the same name as the remote system.

```
                        Change Job Description (CHGJOBD)

Type choices, press Enter.

Job description  . . . . . . . . JOBD          > NETWORK
  Library  . . . . . . . . . . .               >   OSLSYS
Job queue  . . . . . . . . . . . JOBQ            QBATCH
  Library  . . . . . . . . . . .                   QGPL
Job priority (on JOBQ) . . . . . JOBPTY          5
Output priority (on OUTQ)  . . . OUTPTY          5
Print device . . . . . . . . . . PRTDEV          *USRPRF
Output queue . . . . . . . . . . OUTQ          > NETWORK
  Library  . . . . . . . . . . .               >   OSLSYS
Text 'description' . . . . . . . TEXT            'Network utility local jobs.

                           Additional Parameters

User . . . . . . . . . . . . . . USER          > NETWORK

                                                                    More...
F3=Exit   F4=Prompt   F5=Refresh   F10=Additional parameters   F12=Cancel
F13=How to use this display        F24=More keys
```

This job description is by the network utilities when distributing objects from this system.

Press **Enter** to display the description screen then press Enter again to display the Grant Object Authority Screen..

Grant Object Authority Screen

To display this screen, press **Enter** to display the description screen, then press **Enter** again.

Use this screen to grant the required authority to the user profile.

The remote system user profile must have authority to the IBM CRTSAVF command. This command is used by object distribution.

```
                     Grant Object Authority (GRTOBJAUT)

 Type choices, press Enter.

 Object . . . . . . . . . . . . . . OBJ          > CRTSAVF
   Library  . . . . . . . . . . .                >   *LIBL
 Object type  . . . . . . . . . . . OBJTYPE      > *CMD
 Users  . . . . . . . . . . . . . . USER         > NETWORK
                         + for more values
 Authority  . . . . . . . . . . . . AUT          > *USE
                         + for more values
 Authorization list . . . . . . . . AUTL




                                                                     Bottom
 F3=Exit   F4=Prompt   F5=Refresh   F12=Cancel   F13=How to use this display
 F24=More keys
```

Press **Enter** to display the description screen then press Enter again to display the Grant Object Authority Screen 2.

Grant Object Authority Screen 2

To display this screen, press **Enter** to display the description screen, then press **Enter** again.

Use this screen to grant the required authority to the NETWORK user profile, to use the command CHGSYSLIBL.

```
                        Grant Object Authority (GRTOBJAUT)

 Type choices, press Enter.

 Object . . . . . . . . . . . . . . OBJ          > CHGSYSLIBL
   Library  . . . . . . . . . . . .              >   *LIBL
 Object type  . . . . . . . . . . . OBJTYPE      > *CMD
 Users  . . . . . . . . . . . . . . USER         > NETWORK
                           + for more values
 Authority  . . . . . . . . . . . . AUT          > *USE
                           + for more values
 Authorization list . . . . . . . . AUTL


                                                                  Bottom
 F3=Exit   F4=Prompt   F5=Refresh   F12=Cancel   F13=How to use this display
 F24=More keys
```

Press **Enter** to display the description screen, then Press **Enter** to display the Change Directory Entry screen.

Change Directory Entry Screen

To display this screen, press **Enter** to display the description screen, then press **Enter** again.

Use this screen to create or change the directory entry for the user profile.

This contains such information as the user's ID, system name, mailing address, and user profile name.

```
                       Change Directory Entry (CHGDIRE)

Type choices, press Enter.

User identifier:                 USRID
  User ID  . . . . . . . . . . .              > NETWORK
  Address  . . . . . . . . . . .              > NETWORK
User profile . . . . . . . . . . USER         > NETWORK
System name:                     SYSNAME
  System name  . . . . . . . . .              > STUDB30A
  System group . . . . . . . . .
Telephone number 1 . . . . . . . TELNBR1        *SAME

Telephone number 2 . . . . . . . TELNBR2        *SAME

Location . . . . . . . . . . . . LOC            *SAME

Address line 1 . . . . . . . . . ADDR1          *SAME


                                                                     More...
F3=Exit   F4=Prompt   F5=Refresh   F10=Additional parameters   F12=Cancel
F13=How to use this display        F24=More keys
```

Press **Enter** to display the description screen then press **Enter** to display the Change Network Attributes screen.

Change Network Attributes

To display this screen, press **Enter** to display the description screen, then press **Enter** again.

Now you must change the network attributes to reflect the job and message queues created.

Use this screen to change the network attributes.

```
                    Change Network Attributes (CHGNETA)

Type choices, press Enter.

Alert status . . . . . . . . . . ALRSTS        *SAME
Message queue  . . . . . . . . . MSGQ          NETWORK
  Library  . . . . . . . . . . .                 OSLSYS
Output queue . . . . . . . . . . OUTQ          NETWORK
  Library  . . . . . . . . . . .                 OSLSYS
Network job action . . . . . . . JOBACN        *SEARCH
Maximum hop count  . . . . . . . MAXHOP        *SAME
DDM request access . . . . . . . DDMACC        *SAME
  Library  . . . . . . . . . . .


                                                                   Bottom
F3=Exit   F4=Prompt   F5=Refresh   F12=Cancel   F13=How to use this display
F24=More keys
```

Press **Enter** to display the description screen, then press **Enter** to display the Change Network Job Entry screen.

Change Network Job Entry Screen

To display this screen, press **Enter** to display the description screen, then press **Enter** again.

Finally the user NETWORK must have a network job to determine the action to be taken when a user is submitting network jobs.

Use this screen to add or change the network entry.

```
                        Change Network Job Entry (CHGNETJOBE)

 Type choices, press Enter.

 User identifier:                 FROMUSRID
   User ID  . . . . . . . . . . .              > NETWORK
   User ID qualifier  . . . . . .              > NETWORK
 Network job action . . . . . . . ACTION       > *SUBMIT
 User profile . . . . . . . . . . SBMUSER      > NETWORK
 Message queue  . . . . . . . . . MSGQ           NETWORK
   Library  . . . . . . . . . . .                  OSLSYS
 Job queue  . . . . . . . . . . . JOBQ           QBATCH
   Library  . . . . . . . . . . .                  QGPL




                                                                         Bottom
 F3=Exit   F4=Prompt   F5=Refresh   F12=Cancel   F13=How to use this display
 F24=More keys
```

The definition of the local system is now complete. The initial system description is re-displayed. Select **F8** to update the system description.

Add Directory Entry Screen

This only displays for Remote systems.

You must add a distribution directory entry to distribute objects with SNADS to the remote system from the local system. This contains such information as the user ID, system name, mailing address and user profile name:

Use the Add Directory Entry screen to create or change the directory entry:

```
                         Add Directory Entry (ADDDIRE)

 Type choices, press Enter.

 User identifier:                USRID
   User ID  . . . . . . . . . . .              > STUDB30A
   Address  . . . . . . . . . . .              > STUDTOOL
 User description . . . . . . . . USRD         > 'Network Utility from STUDB30A
to STUDTOOL.'
 User profile . . . . . . . . . . USER         > *NONE
 System name:                    SYSNAME
   System name  . . . . . . . . .              > STUDTOOL
   System group . . . . . . . . .
 Telephone number 1 . . . . . . . TELNBR1        *NONE

 Telephone number 2 . . . . . . . TELNBR2        *NONE

 Location . . . . . . . . . . . . LOC            *NONE

                                                                     More...
 F3=Exit   F4=Prompt   F5=Refresh   F10=Additional parameters   F12=Cancel
 F13=How to use this display        F24=More keys
```

Network Manager uses the system user profiles and directory entries to manage the object distribution.

Press **Enter**.

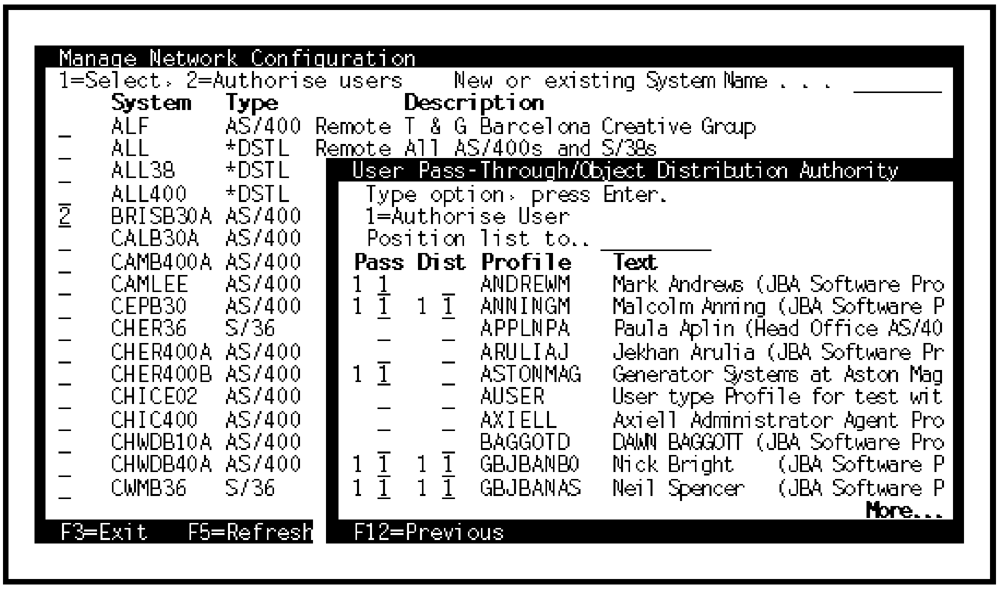
You must add a distribution entry so you can connect from the remote system to the local system.

The definition of the remote system is now complete. The initial system description is re-displayed. Select **F8** to update the system description.

User Pass Through Window

To display this screen, enter 2 alongside a system name from the Manage Network Screen.

Access to distribute objects to and pass through to remote systems is controlled by each user's user profile. For further details about user profiles see the Administration Functions Product Guide.



Fields:

Pass

Enter 1 to allow the user is allowed to pass through to the system.

Dist

Enter 2 in the Dist to allow the user to distribute objects to the system.

Display Network Log

All of the Send Object facilities of Network Manager submit a job to perform the transmission using a job description with the same name as the command, for example, a send library transmission uses a job description SNDLIB.

The submitted job is named after the appropriate library, object, or member being transmitted. The user of the job will be NETWORK regardless of whoever does the submit. Even though the user is NETWORK it will still be displayed by WRKSBMJOB.

When this job has run it will produce two distributions, the DATA and NETWORK job. When the two distributions arrive on the target system, a job will be placed on a job queue of the target system. This job has the same name as the sender job, but the user will be the name of the sending system.

When this job has run the send should be complete. A message is sent to the source system indicating whether the receive was successful or not. This message will arrive on the source system on a message queue whose name matches that of the target system.

Network Manager retains this log of all sent objects.

Display Network Log Screen

To display the details of this log select option 2 from Networking and press **Enter**.

Use this screen to select the object for which you want to display details.

```
NW291                       Network Manager                      System: STUDTOOL
 Display Network Log  - DSPNETLOG
 5=Select Details
   From     To                                   Type/
   System   System    Object      Library        Member        Date     Time    Status
 _ STUDTOOL STUDB30A  XA000       IPGAMP3        *PGM          180691   80412  SS
 _ STUDTOOL STUDB30A  XA005       IPGAMP3        *PGM          180691   80428  SS
 _ STUDTOOL STUDB30A  XA010       IPGAMP3        *PGM          180691   80445  SS
 _ STUDTOOL STUDB30A  XA010CLP    IPGAMP3        *PGM          180691   80455  SS
 _ STUDTOOL STUDB30A  XA011CLP    IPGAMP3        *PGM          180691   80503  SS
 _ STUDTOOL STUDB30A  XA012CLP    IPGAMP3        *PGM          180691   80511  SS
 _ STUDTOOL STUDB30A  XA013CLP    IPGAMP3        *PGM          180691   80519  SS
 _ STUDTOOL STUDB30A  XA200       IPGAMP3        *PGM          180691   80529  SS
 _ STUDTOOL STUDB30A  XA000       IPGAMP3        *PGM          180691   80534  SR
 _ STUDTOOL STUDB30A  QDDSSRC     OSLTEMP        XALIMITS      180691   80600  SS
 _ STUDTOOL STUDB30A  XA005       IPGAMP3        *PGM          180691   80624  SR
 _ STUDTOOL STUDB30A  XA010       IPGAMP3        *PGM          180691   80701  SR
 _ STUDTOOL STUDB30A  XA010CLP    IPGAMP3        *PGM          180691   80748  SR
 _ STUDTOOL STUDB30A  XA011CLP    IPGAMP3        *PGM          180691   80835  SR
 _ STUDTOOL STUDB30A  XA012CLP    IPGAMP3        *PGM          180691   80917  SR
 _ STUDTOOL STUDB30A  XA013CLP    IPGAMP3        *PGM          180691   80956  SR
                                                                                More..
 F3=Exit    F5=Refresh    F12=Previous    F17=Subset
```

Fields:

Untitled

Enter 5 - Select Details to view more details for an entry.

Tip: You can view a subset of this log by selecting **F17** and entering the criteria, such as a sending user to transmissions requested by a specific user.

To view the details of a specific job enter 5 against the required job and press **Enter**.

Display Network Log Details Screen

To display this screen, select a job with option 5 and press **Enter**.

This screen shows the details for the selected job. These details include, the remote and local system ,the users and status of the job.

```
NW291                        Network Manager                    System: STUDTOOL
 Display Network Log  - DSPNETLOG
 From system . . . :  STUDTOOL
 To system . . . . :  STUDB30A

 Object name/Library  XA000      / IPGAMP3
 Type. . . . . . . :  *PGM


 Receiving library :  OSLTEMP

 Sending user. . . :  PEARSONC
 Sending command . :  SNDOBJ

 Status. . . . . . :  SS Sender  Submitted
 Date / Time . . . :  18/06/91     8:04:12
 Job . . . . . . . :  SYSMGR03    PEARSONC     020536
 Message . . . . . :  UNW2100   Job 020536/NETWORK/XA000 submitted to perfor
                      m the send.



 F3=Exit     F5=Refresh     F12=Previous     F17=WRKJOB
```

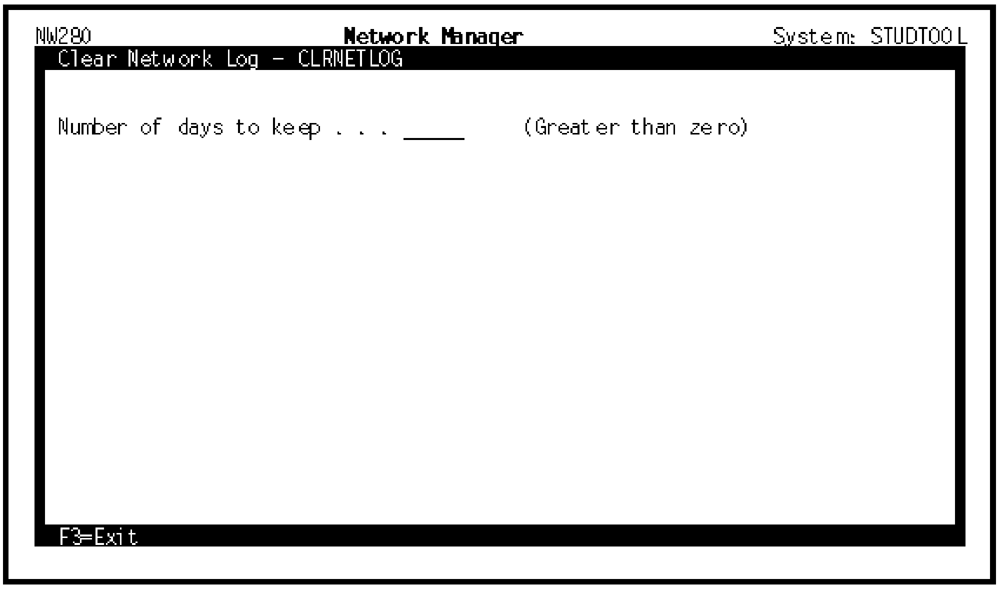
Select **F17=Wrkjob** to work with the submitted job.

Note: If the job is no longer on the system a message is displayed.

Clear Network Log

Select option 3 from Networking and press **Enter**.

Use this screen to specify how much information you want to keep in the log.



Fields:

Number Of Days To Keep

Enter a value greater than zero for the number of days records to keep.

Press **Enter** to submit the clear.

Note: To use the clear Network Log function as part of your daily or weekly system housekeeping, use the command CLRNETLOG.

Manage Pass Through

The pass-through command (PASTHR) calls the IBM supplied command STRPASTHR in order to transfer to a remote system. The soft-coding of the STRPASTHR command considerably simplifies the process.

To display this screen, select option 4 - Manage Pass Through.

```
NW011                     Network Manager                     System: STUDTOOL
 Manage Soft-Coded Pass-Through Options - MNGPASTHR
 Optionally enter 'From' System and 'To' System name . . ________ ________
 1=Select pass-through to maintain, 2=Authorise users
   From     To
 O System   System   Command
 _ STUDTOOL BRISTOOL STRPASTHR RMTLOCNAME(BRISTOOL) VRTCTL(VRTCTL01)
 _ STUDTOOL CALB30A  STRPASTHR RMTLOCNAME(CALB30A) VRTCTL(VRTCTL01)
 _ STUDTOOL SOTN400A STRPASTHR RMTLOCNAME(SOTN400A) VRTCTL(VRTCTL01)
 _ STUDTOOL STUDB30A STRPASTHR RMTLOCNAME(STUDB30A) VRTCTL(VRTCTL03)
 _ STUDTOOL STUDB30B STRPASTHR RMTLOCNAME(STUDB30A) VRTCTL(VRTCTL02)
 _ STUDTOOL STUDDENT STRPASTHR RMTLOCNAME(STUDDENT) VRTCTL(VRTCTL01)
 _ STUDTOOL STUDIOUS STRPASTHR RMTLOCNAME(STUDIOUS) VRTCTL(VRTCTL01)
 _ STUDTOOL STUDPOKR STRPASTHR RMTLOCNAME(STUDPOKR) VRTCTL(VRTCTL01)
 _ STUDTOOL STUDWELD STRPASTHR RMTLOCNAME(STUDWELD) VRTCTL(VRTCTL01)
 _ STUDTOOL WHRDB40A STRPASTHR RMTLOCNAME(WHRDB40A) VRTCTL(VRTCTL01)

                                                                   Bottom
 F3=Exit
```

O

Selection field. Enter one of the following:

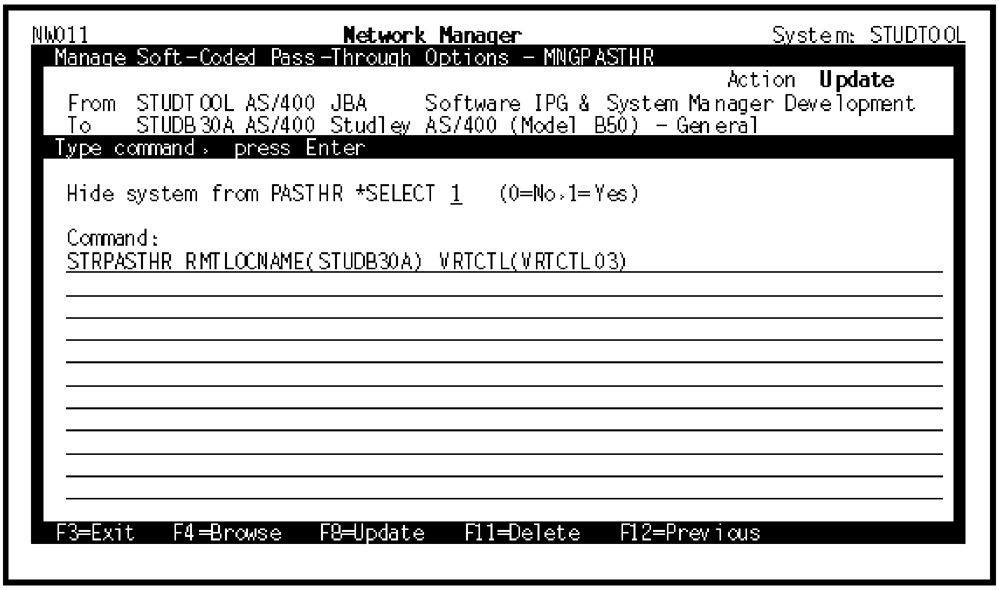
1 - To select the pass-through.

2 - To authorise users do this pass-through.

Add/Maintain Pass Through Command

Enter 1 to select an existing pass-through or
enter a new to/from system pairing.

Use this screen to enter the STRPASTHR command to use.



Fields:

Hide System From Pasthr

If you do not want this system available on the Pass-through System Selection Display, enter Y against this option.

Command

Enter the command required to initiate pass-through to the remote system. Press F4 to use the IBM command prompter and enter the required parameters for the pass-through command. The entered commands are validated and the command string is highlighted if any syntax errors are found.

Functions:

F8

Update the command options.

F11

Delete a pass-through option, press F11 again to confirm.

F16

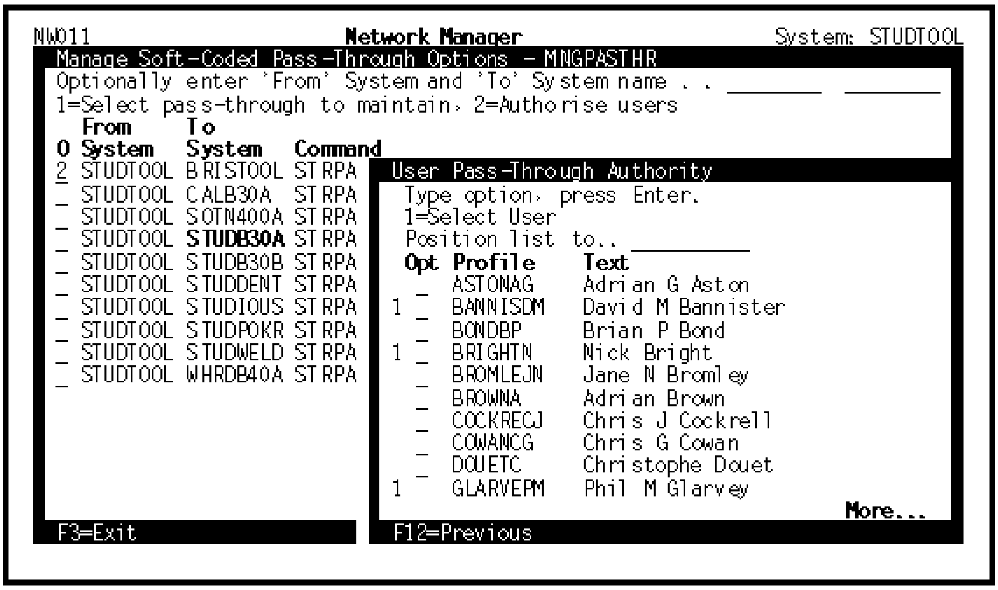
Execute the start pass-through command from this display.

See the Pass-through section for details on initiating pass-through.

User Pass-Through Authority Window

Enter 2 against a system in the initial screen to display this window.

Use this window to authorise users to this pass-through.



Enter 1 against a user to authorise them to pass-through to this system.

Examples Of The Start Pass Through Command

APPC Networks

For adjacent systems in an APPC network:

STRPASTHRRMTLOCNAME(*CNNDEV)

CNNDEV(BRISB30A) VRTCTL(VRTCTL01)

For non-adjacent systems in an APPC network (multiple hops)

STRPASTHRRMTLOCNAME(*CNNDEV)

CNNDEV(SYS1SYS2)VRTCTL(VRTCTL01)

APPN Networks

For adjacent or non-adjacent systems in an APPN network:

STRPASTHRRMTLOCNAME(BRISB30A)

VRTCTL(VRTCTL01)

For an APPN network where automatic sign-on is required:

STRPASTHRRMTLOCNAME(SYS1)

VRTCTL(VRTCTL01)RMTUSER(SALES)RMTPWD(SAL011)

Note: Automatic sign-on must be permitted on the target system, and the specified user profile amended for this option.

Multiple Networks

STRPASTHRRMTLOCNAME(CHIC400)

CNNDEV(AKA400)VRTCTL(VRTCTL01)

This would initiate pass-through to node CHIC400 and then select device AKA400 at the remote system to access an attached network or node via an APPC bridge.

Pass-through

Pass-through can be initiated in three ways:

1. Select option 5 from Networking and select a system from the System Selection Display by entering a 1 against it.
2. Enter the command PASTHR with no parameters. The System Selection Display screen appears. Enter 1 against the required system and press **Enter**.
3. Enter the command PASTHR with the target system name, for example: PASTHRSTUDB30A

If you use method 1 or 2, enter 1 against the required system and press **Enter**.